

DATABASE SYSTEM ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT

SKILL SESSION 5 — LAB RECORD

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Section	7
Course	Database System Engineering and Distributed Backend Development (25CS1302E)
Experiment Title	Bank Transaction Management Using MongoDB Atlas — Aggregation Pipelines
Platform	MongoDB Atlas — Aggregation Pipeline Builder
Database	bankdb

4. Aggregation Queries (Q12 – Q20)

Query 12: Sort Customers by Balance (Descending)

Collection: customers

Stage 1 — \$sort:

```
{ "balance": -1 }
```

The screenshot displays the MongoDB Atlas Aggregation Pipeline Builder interface. On the left, the 'Data Explorer' sidebar shows the 'customers' collection under the 'bankdb' database. The main workspace is titled 'Cluster0 > bankdb > customers' and shows the 'Aggregations' tab with a single stage named '\$sort'. The stage configuration shows the query '{ "balance": -1 }'. Below the configuration, the 'Output preview' section displays a sample of 8 documents sorted by balance in descending order. The documents are as follows:

Document
<pre>{ "_id": "6a7c38fbcfb073ac6a3bc31", "customer_id": 105, "name": "Kiran Kumar", "city": "Guntur", "account_type": "Current", "balance": 65000, "status": "Inactive" }</pre>
<pre>{ "_id": "6a7c38fbcfb073ac6a3bc35", "customer_id": 103, "name": "Arjun Reddy", "city": "Hyderabad", "account_type": "Savings", "balance": 45000, "status": "Active" }</pre>
<pre>{ "name": "Priya Sharma", "city": "Vijayawada", "account_type": "Current", "balance": 125000, "status": "Active" }</pre>
<pre>{ "name": "Arjun Reddy", "city": "Hyderabad", "account_type": "Savings", "balance": 45000, "status": "Active" }</pre>
<pre>{ "name": "Sneha Rao", "city": "Marangal", "account_type": "Saving", "balance": 210000, "status": "Active" }</pre>
<pre>{ "name": "Ravi Kumar", "city": "Hyderabad", "account_type": "Savings", "balance": 85000, "status": "Active" }</pre>

Query 13: Total Transaction Amount (All Transactions)

Collection: transactions

Stage 1 — \$group:

```
{ "_id": null, "total_amount": { "$sum": "$amount" } }
```

The screenshot shows the MongoDB Atlas Data Explorer interface. On the left, the 'Data Explorer' sidebar shows the 'transactions' collection under the 'bankdb' database. The main panel displays the 'Aggregations' tab for the 'transactions' collection. A single aggregation stage named '\$group' is applied. Below the stage, two sample documents are shown: one for a deposit and one for a withdrawal. At the bottom, the query pipeline is shown as a single stage: `1 $group, "total_amount": { "$sum": "$amount" } }`. To the right, the 'Output preview' shows the result of the query: `{ "_id": null, "total_amount": 602000 }`.

Query 14: Average Transaction Amount

Collection: transactions

Stage 1 — \$group:

```
{ "_id": null, "average_amount": { "$avg": "$amount" } }
```

ORGANIZATION

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PROJECT

Project 0

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

admin

bankdb

customers

transactions

local

sample_mflix

transactions

customers

+

Cluster0 > bankdb > transactions

Documents 15

Aggregations

Schema

Indexes 1

Validation

\$group

\$group

Untitled - modified

SAVE

CREATE NEW

EXPORT CODE

Stage 2

\$group

1, "average_amount": { "\$avg": "\$amount" } }

Output preview after \$group stage (Sample of 1 document)

_id: null

average_amount: null

Query 15: Total Amount Grouped by Transaction Type

Collection: transactions

Stage 1 — \$group:

```
{ "_id": "$type", "total_amount": { "$sum": "$amount" } }
```

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PROJECT

Project 0

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

admin

bankdb

customers

transactions

local

sample_mflix

transactions

customers

+

Cluster0 > bankdb > transactions

Documents 15

Aggregations

Schema

Indexes 1

Validation

\$group

Generate aggregation

Untitled - modified

SAVE

CREATE NEW

EXPORT CODE

PREVIEW

Stage 1

\$group

1 e, "total_amount": { "\$sum": "\$amount" }

Output preview after \$group stage (Sample of 2 documents)

customer_id: 101
type: "Deposit"
amount: 25000
mode: "UPI"
date: "2026-08-01"
status: "Success"

customer_id: 101
type: "Withdrawal"
amount: 10000
mode: "ATM"
date: "2026-08-02"
status: "Success"

customer_id: 102
type: "Deposit"
amount: 50000
mode: "NEFT"
date: "2026-08-02"
status: "Success"

Query 16: Number of Transactions per Customer

Collection: transactions

Stage 1 — \$group:

```
{ "_id": "$customer_id", "transaction_count": { "$sum": 1 } }
```

Stage 2 — \$sort:

```
{ "transaction_count": -1 }
```

The screenshot displays the MongoDB Data Explorer interface. On the left, the 'Data Explorer' sidebar shows the 'transactions' collection under the 'bankdb' cluster. The main workspace is titled 'Cluster0 > bankdb > transactions' and shows a query pipeline with two stages: '\$group' and '\$sort'. The '\$group' stage is selected, and its output preview shows three documents with '_id' values 103, 104, and 105, and a 'transaction_count' of 2, 2, and 1 respectively. The '\$sort' stage is also shown, and its output preview shows two documents with '_id' values 106 and 108, and a 'transaction_count' of 2 and 2 respectively. The interface includes buttons for 'View monitoring', 'Visualize', 'Generate aggregation', 'Explain', 'Run', 'PREVIEW', 'STAGES', 'TEXT', and 'W'. The query pipeline is defined as follows:

```
1 { "_id": "$customer_id", "transaction_count": { "$sum": 1 } }
2 { "transaction_count": -1 }
```

Query 17: Join Transactions with Customer Details (\$lookup)

Collection: transactions

Stage 1 — \$lookup:

```
{
  "from": "customers",
  "localField": "customer_id",
  "foreignField": "customer_id",

  "as": "customer_details"
}
```

Stage 2 — \$unwind:

```
"$customer_details"
```

The screenshot displays the MongoDB Atlas Data Explorer interface. On the left, the 'Data Explorer' sidebar shows the 'transactions' collection under the 'bankdb' database. The main workspace is titled 'Cluster0 > bankdb > transactions' and shows a query pipeline with two stages: '\$lookup' and '\$unwind'.

Stage 1: \$lookup

The '\$lookup' stage is configured with the following settings:

- Documents: 15
- Aggregations: 1
- Schema: 1
- Indexes: 1
- Validation: 1

The output preview for Stage 1 shows a sample of 10 documents. The first document is:

```
{
  "from": "customers",
  "localField": "customer_id",
  "foreignField": "customer_id",
  "as": "customer_details"
}
```

The output preview for Stage 1 shows a sample of 10 documents. The first document is:

```
{
  "_id": ObjectId('6a7c3accecfc073ac6a3bc42'),
  "transaction_id": 1001,
  "customer_id": 101,
  "type": "Deposit",
  "amount": 25000,
  "mode": "UPI",
  "date": "2026-08-01",
  "status": "Success",
  "customer_details": Array (1)
}
```

Stage 2: \$unwind

The '\$unwind' stage is configured with the following settings:

- Documents: 15
- Aggregations: 1
- Schema: 1
- Indexes: 1
- Validation: 1

The output preview for Stage 2 shows a sample of 10 documents. The first document is:

```
{
  "_id": ObjectId('6a7c3accecfc073ac6a3bc42'),
  "transaction_id": 1001,
  "customer_id": 101,
  "type": "Deposit",
  "amount": 25000,
  "mode": "UPI",
  "date": "2026-08-01",
  "status": "Success",
  "customer_details": Object (7)
}
```

Query 18: Customer Name + Transaction Amount (Reshape with \$project)

Collection: transactions

Stage 1 — \$lookup:

```
{
  "from": "customers",
  "localField": "customer_id",
  "foreignField": "customer_id",
  "as": "customer"
}
```

Stage 2 — \$unwind:

```
"$customer"
```

Stage 3 — \$project:

```
{
  "_id": 0,
  "transaction_id": 1,
  "customer_name": "$customer.name",
  "city": "$customer.city",
  "type": 1,
  "amount": 1,
  "mode": 1
}
```

The screenshot displays the MongoDB Atlas web interface for a cluster named 'bankab'. The 'Data Explorer' sidebar on the left shows the database structure, including collections 'admin', 'bankab', 'customers', and 'transactions'. The main panel is set to the 'Aggregations' tab for the 'transactions' collection. The aggregation pipeline consists of three stages: '\$lookup', '\$unwind', and '\$project'. The '\$lookup' stage is currently selected and expanded, showing its configuration and a preview of the resulting documents. The preview shows four documents, each with a '_id' field set to 0, and a 'customer' array containing one object with 'name' and 'city' fields. The '\$project' stage is also visible, showing its configuration and a preview of the resulting documents. The preview shows four documents, each with a '_id' field set to 0, and a 'customer' array containing one object with 'name' and 'city' fields. The interface includes various controls such as 'Generate aggregation', 'Run', 'Options', 'Preview', 'Stages', 'Text', and 'Wizard'.

Stage 3 \$project

```

1 {
2   "_id": 0,
3   "transaction_id": 1,
4   "customer_name": "$customer.name",
5   "city": "$customer.city",
6   "type": 1,
7   "amount": 1,
8   "mode": 1
9 }
10

```

Output preview after \$project stage (Sample of 10 documents)

transaction_id	type	amount	mode	customer_name	city
1001	Deposit	25000	UPI	Ravi Kumar	Hyderabad
1002	Withdrawal	10000	ATM	Ravi Kumar	Hyderabad
1003	Deposit	50000	NEFT	Priy	Vijayawada

Query 19: Total Transaction Amount per Customer (with Name)

Collection: transactions

Stage 1 — \$group:

```
{ "_id": "$customer_id", "total_transaction": { "$sum": "$amount" } }
```

Stage 2 — \$lookup:

```
{
  "from": "customers",
  "localField": "_id",
  "foreignField": "customer_id",
  "as": "customer"
}
```

Stage 3 — \$unwind:

```
"$customer"
```

Stage 4 — \$project:

```
{
  "_id": 0,
  "customer_id": "$_id",
  "customer_name": "$customer.name",
  "total_transaction": 1
}
```

Stage 5 — \$sort:

```
{ "total_transaction": -1 }
```

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PROJECT
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Cluster0 > bankdb > transactions

Documents 15 Aggregations Schema Indexes 1 Validation

\$group \$lookup \$unwind \$project \$sort

Generate aggregation + Explain Run Options

Untitled - modified SAVE + CREATE NEW EXPORT CODE

PREVIEW STAGES TEXT WIZARD

Stage 1 \$group

1 total_transaction: { "\$sum": "\$amount" } }

Output preview after \$group stage (Sample of 8 documents)

_id: 102
total_transaction: 65000

_id: 108
total_transaction: 180000

_id: 106
total_transaction: 9

Stage 2 \$lookup

1 {
2 "from": "customers",
3 "localField": "_id",
4 "foreignField": "customer_id",
5 "as": "customer"
6 }
7

Output preview after \$lookup stage (Sample of 8 documents)

_id: 101
total_transaction: 35000
customer: Array (1)

_id: 106
total_transaction: 95000
customer: Array (1)

_id: 108
total_transaction: 1
customer: Array (1)

Stage 3 \$unwind

1 "\$customer"

Output preview after \$unwind stage (Sample of 8 documents)

_id: 103
total_transaction: 35000
customer: Object (7)

_id: 104
total_transaction: 125000
customer: Object (7)

_id: 105
total_transaction: 1
customer: Object (7)

Stage 4 \$project

1 {
2 "_id": 0,
3 "customer_id": "\$_id",
4 "customer_name": "\$customer.name",
5 "total_transaction": 1
6 }
7

Output preview after \$project stage (Sample of 8 documents)

total_transaction: 95000
customer_id: 106
customer_name: "Anjali Singh"

total_transaction: 52000
customer_id: 107
customer_name: "Rahul Verma"

total_transaction: 6
customer_id: 102
customer_name: "Priy"

ORGANIZATION
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PROJECT
Project 0

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Query 20: Complex Bank Transaction Report

Collection: customers

Stage 1 — \$match:

```
{ "status": "Active", "balance": { "$gt": 80000 } }
```

Stage 2 — \$lookup:

```
{
  "from": "transactions",
  "localField": "customer_id",
  "foreignField": "customer_id",
  "as": "transactions"
}
```

Stage 3 — \$unwind:

```
"$transactions"
```

Stage 4 — \$group:

```
{
  "_id": "$customer_id",
  "customer_name": { "$first": "$name" },
  "city": { "$first": "$city" },
  "balance": { "$first": "$balance" },
  "total_transactions": { "$sum": "$transactions.amount" },
  "transaction_count": { "$sum": 1 }
}
```

Stage 5 — \$sort:

```
{ "total_transactions": -1 }
```

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

admin

bankdb

customers

transactions

local

sample_mflix

transactions

customers

+

Cluster0 > bankdb > customers

View monitoring Visualize Your Data

Documents 8

Aggregations

Schema

Indexes 1

Validation

\$match

\$lookup

\$unwind

\$group

\$sort

Generate aggregation Explain Run Options

Untitled - modified

SAVE

+ CREATE NEW

EXPORT CODE

PREVIEW

STAGES

TEXT

WIZARD

Stage 1 \$match

1 s: "Active", "balance": { "\$gt": 80000 } }

Output preview after \$match stage (Sample of 6 documents)

_id: ObjectId('6a7c3854ecfb073ac6a3bc31')

customer_id: 101

name: "Ravi Kumar"

city: "Hyderabad"

account_type: "Savings"

balance: 85000

status: "Active"

_id: ObjectId('6a7c3881ecfb073ac6a3bc33')

customer_id: 102

name: "Priya Sharma"

city: "Vijayawada"

account_type: "Current"

balance: 125000

status: "Active"

_id: ObjectId('6a7c38e6ecfb073ac6a3bc35')

customer_id: 104

name: "Sneha Rao"

city: "Marangal"

account_type: "Savin"

balance: 210000

status: "Active"

Stage 2 \$lookup

1 {
2 "from": "transactions",
3 "localField": "customer_id",
4 "foreignField": "customer_id",
5 "as": "transactions"
6 }
7

Output preview after \$lookup stage (Sample of 6 documents)

_id: ObjectId('6a7c3854ecfb073ac6a3bc31')

customer_id: 101

name: "Ravi Kumar"

city: "Hyderabad"

account_type: "Savings"

balance: 85000

status: "Active"

transactions: Array (2)

_id: ObjectId('6a7c3881ecfb073ac6a3bc33')

customer_id: 102

name: "Priya Sharma"

city: "Vijayawada"

account_type: "Current"

balance: 125000

status: "Active"

transactions: Array (2)

_id: ObjectId('6a7c38e6ecfb073ac6a3bc35')

customer_id: 104

name: "Sneha Rao"

city: "Marangal"

account_type: "Savin"

balance: 210000

status: "Active"

transactions: Array

Stage 3 \$unwind

1 "\$transactions"

Output preview after \$unwind stage (Sample of 10 documents)

_id: ObjectId('6a7c3854ecfb073ac6a3bc31')

customer_id: 101

name: "Ravi Kumar"

city: "Hyderabad"

account_type: "Savings"

balance: 85000

status: "Active"

transactions: Object (8)

_id: ObjectId('6a7c3854ecfb073ac6a3bc31')

customer_id: 101

name: "Ravi Kumar"

city: "Hyderabad"

account_type: "Savings"

balance: 85000

status: "Active"

transactions: Object (8)

_id: ObjectId('6a7c38e6ecfb073ac6a3bc35')

customer_id: 102

name: "Priya Sharma"

city: "Vijayawada"

account_type: "Curre"

balance: 125000

status: "Active"

transactions: Object

The screenshot displays the Databricks Data Explorer interface. On the left sidebar, the cluster selection tree shows 'Cluster0' as the active cluster, with databases 'bankdb' and 'customers' listed under it. The main workspace area is titled 'Data Explorer' and shows a pipeline configuration for 'Cluster0 > bankdb > customers'. The pipeline consists of two stages:

- Stage 4 (\$group)**: This stage performs an aggregation. The configuration bar shows '\$match', '\$lookup', '\$unwind', '\$group', and '\$sort'. Below the configuration, there are buttons for 'Generate aggregation', 'Explain', 'Run', and 'Options'. The output preview for Stage 4 shows three sample documents, each representing a different customer (_id: 104, 106, 107) with their respective details like name, city, balance, and transaction counts.
- Stage 5 (\$sort)**: This stage sorts the data based on the 'total_transactions' field. The configuration bar shows '\$total_transactions': -1. The output preview for Stage 5 shows three sample documents, which are now sorted by their total transaction count (e.g., _id: 108 has 320000 transactions, while _id: 104 has 2).

The bottom of the interface includes navigation controls and a search bar.